M2 | Mid-submission

Lloyds Green Customer Insights Platform | 11 October 2026 | Current progress and experimental evidence

1. Current approach and development since M1

We are implementing a customer-level sustainability insights platform. Its core flow is transaction categorisation → Customer 360 feature engineering → four product-propensity models → use-case screening → deterministic sustainability-impact calculation → grounded natural-language explanations and customer/business dashboards. ML predicts propensity; rule-based screening does not claim bank approval; the LLM must explain only computed outputs.

Work completed during the current milestone: Established a reproducible Python/scikit-learn pipeline, trained and saved four propensity pipelines plus a transaction classifier, generated evaluation artifacts, and added customer-disjoint data splits and artifact audits. A uv-based Python environment is configured. Backend execution and integrated UI testing remain in progress; a FastAPI dependency/interpreter mismatch was encountered and installation in the project environment is being addressed.

Data enrichment: Processed ~8 GB of Low Carbon London smart-meter files into household electricity statistics (5,561 metered households: estimated annual mean 3,729.9 kWh; median 3,043.0 kWh). Used these consumption distributions to inform 5,000 fictional banking-customer profiles. These are synthetic financial customers and simulated product take-up outcomes, not matched real bank records. Annualisation and proxy day-coverage assumptions are documented limitations.

2. Experiments run and results

Data preparation and splitting. Synthetic cohort: 5,000 customers; 368,323 transactions. Customer-disjoint training/validation/test partitions: 3,500 / 750 / 750. Audit found zero overlapping customers, zero orphan records and no target labels inside feature columns.

Experiment A — transaction categorisation. TF-IDF (unigrams/bigrams) + Logistic Regression against a rules baseline, measured with macro-averaged F1 over eight sustainability categories.

Method	Validation macro F1	Held-out test macro F1
TF-IDF + Logistic Regression	0.9808	0.9803
Rule-based baseline	0.9177	0.9181

The ML classifier improves test macro F1 by 0.0622 absolute (6.22 percentage points). Results are limited to synthetic merchant descriptions; merchant templates appear across customer partitions, so performance on unfamiliar real merchants remains unverified.

Experiment B — propensity modeling. For each required product, compared Logistic Regression and Random Forest. Chose the algorithm with lower validation Brier score (probability error), without using the test set for selection.

Product	Selected algorithm	Held-out test ROC-AUC
GP01 · EV Finance	Logistic Regression	0.7508
GP02 · Green Home Loan	Logistic Regression	0.8147
GP03 · Solar Finance	Logistic Regression	0.9009
GP04 · Paperless Banking	Random Forest	0.8657

3. Follow-up experiments and testing progress

Experiment C — decision thresholds. A default probability cut-off of 0.50 yielded poor recall for rare positive labels. Product-specific thresholds were selected on validation data by maximizing F1, then evaluated on held-out test customers. No model retraining was needed for this comparison.

Product	Threshold	Test precision	Test recall	Test F1
GP01 · EV Finance	0.24	0.3551	0.4298	0.3889
GP02 · Green Home	0.10	0.1983	0.9333	0.3271
GP03 · Solar Finance	0.23	0.2800	0.4773	0.3529
GP04 · Paperless	0.28	0.4318	0.8702	0.5772

All four held-out F1 values improved relative to threshold 0.50 (respectively 0.0000, 0.0000, 0.2222, 0.0149). The trade-off is substantial false-positive volume, notably GP02 (353 / 750 test customers flagged, precision 19.83%). Thus propensity ranking and institutional screening must remain distinct from a lending decision.

Artifact and robustness checks. Verified disjoint customer splits, model feature schemas, serialized pipeline loading, training-only preprocessing statistics, saved predictions, and three missing-data scenarios. A strict exact-equality assertion initially failed for model/reloaded probabilities and calibration metrics due to differences of approximately 10^-16; a tolerance-aware verification comparison is being finalized. A diagnostic audit reached the final checks, but permanent verification with all meaningful assertions reinstated must be rerun.

Current integration status. Trained model files and metric reports have been produced. The inference module exposes product propensities, rule-trigger evidence, review/exclusion statuses and data-quality warnings. Selected classification thresholds are separately saved and require integration into the business prioritization layer. The FastAPI startup encountered a missing fastapi dependency under system Python 3.13; project-environment installation/startup requires confirmation. No fully working frontend-backend demonstration is claimed yet.

4. Plan for remaining time

Priority 1 — reliable execution. Install backend dependencies under the uv virtual environment; start API with python -m uvicorn; restore rigorous tolerance-aware artifact verification; preserve current models and reports.

Priority 2 — finish the core user journey. Wire trained models and validation-selected thresholds to backend APIs, keep product-use-case screening and bank eligibility separate, connect deterministic carbon-impact estimates and grounded explanation fallbacks, and finish customer/business dashboard flows.

Priority 3 — end-to-end validation. Test representative customers, missing/conflicting data, excluded recommendations, API responses, model loading, and frontend display. Verify calculated quantities and unsupported-claim handling; keep synthetic and annualised-LCL data limitations visible.

Priority 4 — submission. Finalize README/run commands, evaluation and architecture documentation, responsible-AI notes, a reproducible demo path and required trained artifacts; push all project code to main after checking repository status. Exclude the ~8 GB raw dataset and any secrets from Git.

Evidence boundary: This report covers experiments and milestones verified in the available project outputs. Changes since M1 are described as work completed in this interval; an M1 document was not provided for a line-by-line comparison. All ML scores reported above concern simulated labels, not real-world deployment accuracy.